Christopher Loorkhoor

832-870-7134 – cloorkhoor@outlook.com - www.linkedin.com/in/cloorkhoor

Summary:

Chemical Engineering undergraduate with research experience in heterogeneous catalysis, CO_2 adsorption, process simulation, and computational modeling. Experienced in fixed-bed reactor operation, experimental characterization, and engineering data analysis using Python and Aspen Plus.

Education:

University of Houston, Cullen College of Engineering – Houston, Texas Expected Graduation: **May 2028**
Bachelor of Science in Chemical Engineering

Engineering Experience & Projects:

Ni–Ce Materials | CO_2 Adsorption & Characterization **August 2025 – Present**

- Operated fixed-bed CO_2 adsorption systems with regeneration cycles, using controlled gas flows and online **mass spectrometry** to evaluate sorbent performance.
- Refined reactor pretreatment, regeneration, and breakthrough-curve integration procedures to improve the consistency of CO_2 uptake measurements.
- Designing comparative adsorption experiments for nickel cerium hydroxide and oxide sorbents, supported by **XRD and BET characterization**

Computational Modeling | UH iGEM **June 2026 – August 2026**

- Used **Python and COBRApy** to model citrate synthase variants across 108 conditions, predicting 74% higher average α-ketoglutarate production with R163L versus native GltA.
- Compared native GltA, R163L, and feedback-insensitive CggltA to evaluate how enzyme regulation changed predicted TCA-cycle flux.

Steam Methane Reformer (SMR) Simulation | Aspen Plus **September 2025 – December 2025**

- Built an **Aspen Plus** SMR flowsheet using RStoic reactors, flash drums, mixers, and separators to model reforming and water-gas-shift conversion.
- Designed a five-unit process producing approximately **1.63×10^5 kg/day** of **99.99% H_2** using reforming and water-gas-shift reactions over nickel catalyst.

Gallium/ZSM-5 Catalysis | Fixed-Bed Reactor Testing **June 2025 – August 2025**

- Conducted **fixed-bed reactor tests** with ethanol/ethylene feeds over gallium/ZSM-5 catalysts, increasing aromatic selectivity **6–9%** for sustainable jet fuel applications.
- Used **XRD and BET** to assess catalyst phase, surface area, and porosity, identifying gallium oxides, 290 m^2/g surface area, and a Type IV mesoporous isotherm.

Work Experience:

Teaching Assistant - University of Houston **August 2025 – Present**

- Developed a **Python-based scheduling workflow** for **300+ students and 10 faculty members**, improving efficiency and reducing coordination errors.
- Engage directly with students to answer questions and provide course guidance while coordinating attendance, email communication, and final-exam logistics.

Honors, Awards & Certifications:

Stutts Research Program Scholar August 2025
Summer Undergraduate Research Fellowship June 2025
Dean's List January 2025

AIChE/CCPS SAChE - 6 Process Safety Certificates
Process Safety, Hazard Recognition, HIRA, PHA/HAZOP/LOPA, Process Risk Management Fundamentals | 2026

Skills:

Engineering & Simulation: Aspen Plus, fixed-bed reactor operation, gas handling/flow control
Programming & Data Analysis: Python, MATLAB, COBRApy, Excel, regression, curve fitting, numerical integration
Laboratory & Characterization: Gas chromatography (GC), mass spectrometry (MS), XRD, BET

Interests:

Sprinting & Track, Bass Guitar & Double Bass, Strength Training, Web Development